

Adaptive Token Refinement in Long-Tailed Large Vision-Language Models Fine-Tuning

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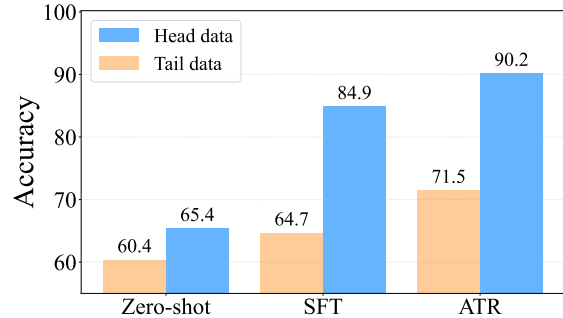
Abstract

While large vision-language models (LVLMs) have shown remarkable adaptability to downstream applications, their fine-tuning process remains susceptible to bias under long-tailed data. Compared to zero-shot scenarios, fine-tuning LVLMs on imbalanced datasets often yields limited performance improvements on tail data. This is because LVLMs tend to rapidly overfit the head data at an early fine-tuning stage, thereby impairing the learning of the tail data while simultaneously failing to exploit their quantitative advantage. Furthermore, in many downstream LVLM scenarios, quantified long-tailed prior knowledge of data distribution is often unavailable, significantly limiting the applicability of traditional long-tailed techniques that rely heavily on such information. To address these issues, we propose the Adaptive Token Refinement (ATR), a novel framework that adaptively refines the learning process of LVLMs under long-tailed data. Specifically, ATR consists of two token-level operations applied to output and input tokens, respectively: 1) a bounded adaptive loss that dynamically filters and reweights output tokens to mitigate overfitting on head data, and 2) a visual token mask strategy that augments the probability paths of input tokens to enhance long-tailed performance. Extensive experiments demonstrate that ATR consistently enhance both performance and generalization for long-tailed LVLMs fine-tuning.

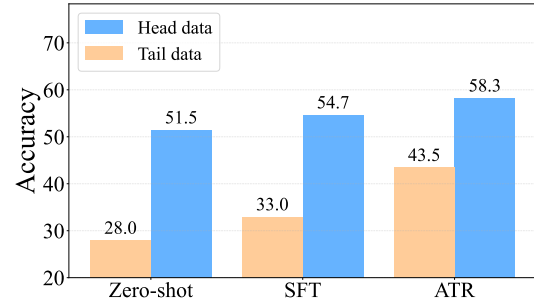
1. Introduction

Recently, large vision-language models (LVLMs), which equip large language models with vision encoders, have

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(a) The performance of head and tail data on ImageNet-LT.



(b) The performance of head and tail data on Place-LT.

Figure 1. The long-tailed performance of Qwen2.5-VL-7B (Team, 2025) training with LoRA. “SFT” means training on full dataset. Our proposed ATR consistently improve the performance of LVLMs fine-tuning under long-tailed data in both scenarios.

demonstrated remarkable capabilities when fine-tuned for downstream tasks (Team, 2025; Liu et al., 2024a). However, their application to real-world scenarios, such as e-commerce (Zhou et al., 2025) and medical diagnosis (Hu et al., 2024), remains challenging due to the presence of long-tailed distribution in the fine-tuning data. In such case, a few head knowledge encompass most of the data, while a large number of tail knowledge is represented by very few samples (Zhang et al., 2023; Liu et al., 2019). Therefore, as illustrated in Fig. 1, fine-tuning LVLMs on such a long-tailed data distribution results in only limited performance improvements on tail data. This is because LVLMs tend to rapidly overfit the head data at an early fine-tuning stage, impairing the learning of the tail data while simultaneously

failing to exploit their quantitative advantage.

Although there have been many efforts for visual recognition under long-tailed data (Miao et al., 2024a;b), they typically rely on explicit prior knowledge of the data distribution, such as the number of samples per class or pre-defined imbalance ratios. Unfortunately, in many downstream LVLMs applications (e.g. visual question answering (Zhang et al., 2025; Dai et al., 2023)), such information is often unavailable or difficult to quantify accurately, significantly limiting the applicability of existing long-tailed learning approaches. Recently, the emergence of vision-language models like CLIP (Radford et al., 2021) has shown promising results in learning robust representations from long-tailed data by leveraging the alignment between visual and textual modalities (Jiang et al., 2025; Shi et al., 2023). However, current long-tailed learning techniques are primarily designed for classification tasks and cannot be applied to generation scenarios. The long-tailed problem in LVLMs presents unique challenges due to their next token prediction paradigm for generation task, which still remains under-explored.

To address these challenges, we propose adaptive token refinement (ATR), a novel framework that adaptively refines the learning process of LVLMs under long-tailed data distributions. ATR employs two complementary token-level operations: a bounded adaptive loss (BAL) applied to output tokens and a visual token mask strategy (VTM) for input tokens. Specifically, BAL dynamically filters high-confidence tokens to reduce redundancy in well-learned head tokens, while reweighting low-confidence tokens to prevent the sharp optimization of spurious correlations. This strategy better leverages the robust knowledge from pre-trained models, promoting balanced learning that mitigates overfitting and preserves generalization. Furthermore, VTM randomly masks low-entropy visual tokens during training, thereby augmenting the probability paths of the input tokens. This process reduces LVLMs reliance on simplistic cues and diversifies the learning signal, thus enhancing generalization to tail data. By operating at a fine-grained token level, ATR facilitates effective long-tailed learning in LVLMs fine-tuning across diverse scenarios. Our contributions can be summarized as follows:

- We provide a systematic analysis of the long-tailed problem in LVLMs fine-tuning, revealing that these models tend to rapidly overfit the head data while simultaneously impairing the learning of the tail data.
- To address these issues, we propose Adaptive Token Refinement (ATR), the first framework designed to effectively address the long-tailed problem in LVLMs fine-tuning. ATR adaptively refines the learning process of LVLMs by dynamically filtering and reweighting output tokens while masking low-entropy input

tokens, thereby effectively mitigating overfitting on head data and improving generalization for tail data.

- Comprehensive experiments on large-scale long-tailed benchmarks and visual question answering tasks show that ATR significantly improves the performance and generalization of LVLMs under long-tailed data.

2. Related Work

Large Vision-Language Models Fine-tuning. Large vision-language models (LVLMs) have demonstrated remarkable capabilities across diverse downstream applications by capturing rich semantic relationships between visual and textual modalities through large-scale pre-training (Liu et al., 2023). This success has spurred the development of both proprietary models (e.g., GPT-4o (Hurst et al., 2024), Gemini (Team et al., 2023)) and open-source models (e.g., LLaVA (Liu et al., 2024a), Qwen-VL (Team, 2025)). The prevailing LVLM architecture integrates a vision encoder (e.g., SigLip (Tschannen et al., 2025)), a visual projection module, and a large language model as the reasoning engine. To enable efficient task-specific adaptation, parameter-efficient fine-tuning (PEFT) methods, such as LoRA (Hu et al., 2022) and Adapter (Rücklé et al., 2021), have become standard practice, achieving comparable performance to full fine-tuning at significantly reduced computational cost. Despite these architectural advances, LVLMs often exhibit suboptimal performance when fine-tuned on long-tailed datasets. This is because the standard cross-entropy loss, commonly used for PEFT, is ineffective for imbalanced data distribution (Lin et al., 2017; Menon et al., 2020), resulting in the model being biased towards the head data. Some recent methods have attempted to explore alternatives to the standard cross-entropy loss for effective learning. FisherSFT (Deb et al., 2025) enhances statistical efficiency by selecting an informative subset of training examples. MinorSFT (Xie et al., 2024) introduces a dynamic sample-level coefficient to prevent the fine-tuned LLM from deviating too much compared to the reference model. However, these methods are not design to handle imbalanced scenarios and consequently struggle to perform well under long-tailed data distributions. Therefore, this work aims to bridge this critical gap, facilitating the more robust and practical deployment of LVLMs.

Long-Tailed Learning. Long-tailed learning aims to improve the accuracy of tail data with the least influence on the head data. In recent years, long-tailed learning has been developed extensively, including three popular paradigms: class-level semantic enhancement (Liu et al., 2019; Menon et al., 2020), re-sampling strategies (Wallace et al., 2011; Chawla et al., 2002; Zhao et al., 2025), and LLM-empowered text description (Tian et al., 2022;

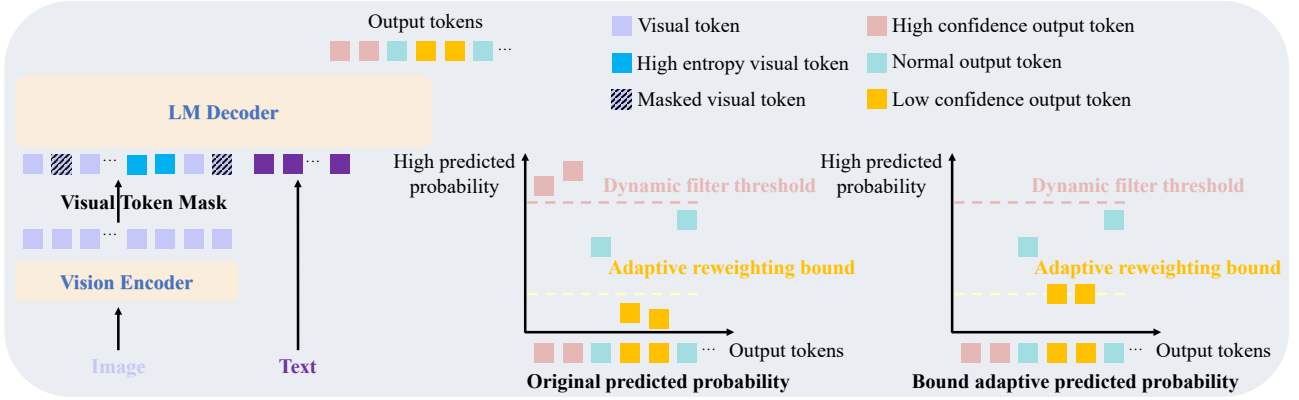


Figure 2. Overview of the proposed Adaptive Token Refinement (ATR) framework, which consists of two main components: (1) Bounded Adaptive Loss, which dynamically filters high-confidence output tokens with an adaptively increasing threshold and reweights low-confidence tokens with a fixed lower bound to prevent overfitting on head data while enhancing tail class learning; (2) Visual Token Mask strategy, which computes token entropy and randomly masks low-entropy visual tokens during training to reduce LVLMs’ reliance on simplistic cues and diversify learning signals, thereby improving tail class generalization.

Jia et al., 2025). Class-level semantic enhancement improves tail class understanding through specialized loss functions (e.g. class-balanced loss (Lin et al., 2017), logit adjustment (Menon et al., 2020)) or architectural modifications (e.g., two-stage decoupling (Alshammari et al., 2022)). Re-sampling strategies balance data distributions through oversampling tail classes (Chawla et al., 2002) or under-sampling (Wallace et al., 2011) head classes, often augmented by external sample generation via image manipulation (Zhao et al., 2025) or text-to-image synthesis (Zhao et al., 2024). LLM-empowered text augmentation leverages large language models to generate fine-grained descriptive prompts to enhance vision-language alignment (Jia et al., 2025). While these methods have achieved success in long-tailed recognition, they face critical limitations when applied to LVLMs. First, they typically require explicit prior knowledge of data distribution, such as per-class sample counts or predefined imbalance ratios, which is often unavailable or difficult to quantify in complex downstream tasks like visual question answering (Zhang et al., 2025; Dai et al., 2023) that long-tailed biases are entangled across multiple dimensions (reasoning types, question formats, linguistic patterns). Second, although recent vision-language models like CLIP (Radford et al., 2021) have shown promising results in learning robust long-tailed representations via cross-modal alignment, existing long-tailed techniques are primarily designed for classification tasks and cannot be applied to generation scenarios where LVLMs operate via the next token prediction paradigm. Recently, some efforts (Yang et al., 2024; Song et al., 2025) have begun exploring the long-tailed problem in LVLMs pre-training data, and address this issue through data resampling strategies. However, the fine-tuning of LVLMs under long-tailed distributions remains significantly underexplored, particularly in the absence of long-tailed distribution priors. This work

bridges this critical gap by proposing a adaptive token-level framework that effectively address the long-tailed problem in diverse LVLM fine-tuning scenarios.

3. Method

Preliminaries. We consider the standard architecture of large vision-language models (LVLMs), which comprises a visual encoder that extracts and projects image features into a sequence of visual tokens, and a large language model backbone that processes both visual and textual information for auto-regressive generation. Formally, let $\mathbf{V} = \{v_1, v_2, \dots, v_N\}$ denote the full set of N visual tokens extracted from the input image, where each token $v_t \in \mathbb{R}^d$ corresponds to a local region of the image. The language model backbone processes the concatenated sequence of visual tokens $\mathbf{V}$ and text tokens $\mathbf{T}$ as input, generating the output token sequence $\mathbf{Y} = \{y_1, y_2, \dots, y_L\}$, where L represents the target sequence length. During standard fine-tuning for downstream tasks, the training objective minimizes the cross-entropy loss over all generated tokens:

$$\mathcal{L}_{\text{CE}} = \mathbb{E}_{(\mathbf{V}, \mathbf{T}) \sim \mathcal{D}} \left[-\frac{1}{L} \sum_{t=1}^L \log P_t \right], \quad (1)$$

where $\mathcal{D}$ represents the fine-tuning dataset, and P_t denotes the predicted probability for the ground-truth token y_t^* at position t :

$$P_t = P(y_t^* | \mathbf{V}, \mathbf{T}, y_{<t}). \quad (2)$$

Here, $P(\cdot)$ represents the softmax-normalized logits, and $y_{<t}$ denotes the sequence of previously generated tokens up to position $t-1$. Traditional long-tailed learning approaches in classification settings typically assume a single output token representing the class label and leverage ground-truth

class frequencies as prior knowledge for distribution-aware optimization. However, LVLMs present fundamental challenges to this paradigm. Their output consists of multi-token sequences $\mathbf{Y}$ with variable lengths and complex semantic dependencies, while input text prompts $\mathbf{T}$ exhibit substantial diversity in format, style, and content. This complexity renders traditional frequency-based distribution quantification impractical, as long-tailed biases manifest across multiple intertwined dimensions.

3.1. Overview of Our Proposed ATR Framework

To address the long-tailed problem in LVLMs fine-tuning, we propose a novel adaptive token refinement (ATR) framework. As illustrated in Figure 2, ATR introduces two core fine-grained token-level components: the bounded adaptive loss (BAL) applied to the output tokens $\mathbf{Y}$, and the visual token mask (VTM) strategy applied to the input visual tokens $\mathbf{V}$. BAL mitigates overfitting on well-learned head concepts and encourages robust learning by avoiding sharp optimization that could lead to spurious correlation. VTM forces the model to develop more robust visual representations that improve tail data performance. Below we introduce each component in detail.

3.2. Bounded Adaptive Loss

The standard cross-entropy loss treats all output tokens equally, which is suboptimal under long-tailed distributions. To address this, we propose BAL, which consists of two modules: dynamic filtering of high-confidence tokens and adaptive reweighting of low-confidence tokens, to respectively reduce the focus on well-learned concepts and mitigate sharp optimization for head concepts, thereby promoting a more balanced learning process. Below we elaborate on how these two modules can help balanced learning.

Dynamic Filtering of High-Confidence Tokens. Due to their powerful pre-training and excessive fine-tuning optimization, LVLMs can rapidly become overconfident in tokens associated with frequent head concepts, leading to irreversible overfitting on head data and impaired learning on tail data. To mitigate this issue, BAL employs a dynamic confidence threshold τ that adaptively filters high-confidence tokens from loss computation. The threshold τ is initialized to a relatively low value and gradually increases throughout training, ensuring that the filtering mechanism becomes more selective as the model’s overall competence improves. When the predicted probability for a ground-truth token exceeds the current threshold, the corresponding loss contribution is excluded from the optimization objective. This selective filtering is formalized as:

$$\mathcal{L}_{\text{filter}} = \mathbb{E}_{(\mathbf{V}, \mathbf{T}) \sim \mathcal{D}} \left[-\frac{1}{L} \sum_{t=1}^L \mathbb{I}_{\tau}[P_t] \cdot \log P_t \right], \quad (3)$$

where $\mathbb{I}_{\tau}[\cdot]$ represents an indicator function defined as:

$$\mathbb{I}_{\tau}[P_t] = \begin{cases} 0 & \text{if } P_t > \tau \\ 1 & \text{otherwise} \end{cases} \quad (4)$$

The dynamic threshold update mechanism ensures adaptive filtering of well-learned tokens based on learning progress. The threshold τ is updated as:

$$\tau \leftarrow \tau + r \cdot \mathbb{I}_{\tau}[P_t] \left(\frac{P_t}{\tau} \right)^{\gamma}, \quad (5)$$

where r controls the incremental update rate and γ modulates the influence of individual token predictions on threshold adjustment. The update magnitude is proportional to the proximity between a token’s predicted confidence and the current threshold, enabling automatic calibration that reflects the model’s evolving competence. This approach effectively prevents overfitting on dominant patterns while redirecting optimization focus toward improving performance on challenging data.

Adaptive Reweighting of Low-Confidence Tokens.

While increasing loss contribution for low-confidence samples represents a useful approach in traditional imbalanced learning, such sharp optimization can be counterproductive in LVLM fine-tuning scenarios. Sharp optimization on difficult samples may cause powerful pre-trained models to develop biases toward specific spurious patterns, increasing overfitting risk and hindering the model’s ability to leverage its pre-trained knowledge for robust task understanding. To address this challenge, BAL employs a principled loss upper-bound mechanism that adaptively reweights low-confidence tokens to prevent excessive optimization while promoting better utilization of pre-trained knowledge. The reweighting is implemented through a bounded weighting factor w_t for each ground-truth token:

$$w_t = \min \left(1, \frac{\ell_{\max}}{-\log P_t + \epsilon} \right), \quad (6)$$

where $\ell_{\max}$ represents the maximum allowable loss value, and ϵ is a small constant that prevents numerical instability from division by zero. This formulation ensures that tokens with extremely low-confidence do not dominate the loss computation, maintaining training stability while still providing appropriate emphasis on challenging samples.

The complete bounded adaptive loss combines both filtering and reweighting mechanisms:

$$\mathcal{L}_{\text{BAL}} = \mathbb{E}_{(\mathbf{V}, \mathbf{T}) \sim \mathcal{D}} \left[-\frac{1}{L} \sum_{t=1}^L \mathbb{I}_{\tau}[P_t] \cdot w_t \cdot \log P_t \right]. \quad (7)$$

This formulation ensures that only tokens below the confidence threshold contribute to loss computation, with their

contributions appropriately scaled to prevent sharp optimization. This dual mechanism allows BAL to effectively balance the learning process between head and tail data.

3.3. Visual Token Mask Strategy

To learn robust and diverse visual representations for underrepresented tail data, ATR incorporates the visual token mask (VTM) strategy to augment the probability paths of input tokens. VTM quantifies token informativeness via token entropy, selectively masking low-entropy tokens while preserving high-entropy tokens that carry rich visual information. This process reduces reliance on simplistic visual cues and improves generalization on tail data across diverse visual scenarios. Formally, given a sequence of visual tokens $\mathbf{V} = \{v_1, v_2, \dots, v_N\}$, where N is the number of visual tokens. VTM computes the token-level generation entropy for each visual token $v \in \mathbf{V}$ as:

$$H(v) = - \sum_{i=1}^{|\mathcal{V}|} p_i \log_2 p_i, \quad (8)$$

where $|\mathcal{V}|$ represents the vocabulary size, and p_i denotes the predicted probability of visual token v for each vocabulary token i . Once the token entropy is computed, VTM randomly masks a proportion of the low-entropy visual tokens, which typically correspond to predictable or redundant information. Let $H_k(\mathbf{V})$ denote the k -th percentile of highest token entropy within the visual tokens set $\mathbf{V}$. The set of tokens to be masked, $\mathbf{V}'$, is constructed by evaluating each token $v \in \mathbf{V}$ as follows:

$$\mathbf{V}' = \{v \mid H(v) < H_k(\mathbf{V}) \text{ and } \text{Bernoulli}(m) = 1\}, \quad (9)$$

where $\text{Bernoulli}(m)$ denotes a Bernoulli random variable that returns 1 with probability m . This formulation ensures that a token is masked only if it is a low-entropy token (below the k -th percentile) and is randomly selected with probability m . The final set of visual tokens used during training is obtained by removing the masked tokens:

$$\mathbf{V}_{\text{VTM}} = \mathbf{V} - \mathbf{V}'. \quad (10)$$

This masking strategy provides a lightweight and effective alternative that avoids the need to compute full attention scores, making it directly compatible with accelerated training methods such as FlashAttention.

The overall training objective of the ATR framework for LVLM fine-tuning is formulated as:

$$\mathcal{L}_{\text{ATR}} = \mathbb{E}_{(\mathbf{V}_{\text{VTM}}, \mathbf{T}) \sim \mathcal{D}} \left[-\frac{1}{L} \sum_{t=1}^L \mathbb{I}_\tau[P_t] \cdot w_t \cdot \log P_t \right], \quad (11)$$

where P_t is defined in Eq. 2, $\mathbb{I}_\tau[\cdot]$ is defined in Eq. 4, and w_t is defined in Eq. 6. By refining both the input and output

sides of the LVLM fine-tuning process, the ATR framework provides a robust and effective solution to enhance long-tail performance for LVLMs fine-tuning.

4. Experiments

4.1. Experimental Setup

Datasets. To comprehensively evaluate our proposed ATR, we employ two distinct categories of datasets: long-tailed image classification benchmarks and visual question answering (VQA) datasets, which allows us to assess ATR’s effectiveness across scenarios with explicit class imbalances and those with more complex, entangled long-tailed biases. **1) Long-Tailed Classification Dataset.** We use ImageNet-LT (Liu et al., 2019) and Place-LT (Liu et al., 2019), two large-scale benchmarks with quantifiable class imbalances that enable clear analysis of performance across head, medium, and tail subsets. ImageNet-LT has 115.8K images from 1,000 classes, with a maximum of 1,280 and a minimum of 5 images per class. Places-LT contains 62.5K images from 365 classes, from a maximal 4,980 to a minimum of 5 images per class. Furthermore, to evaluate the zero-shot generalization capability of our proposed ATR during LVLM fine-tuning, we use ImageNet-LT and Places-LT as cross-dataset test sets. **2) VQA Dataset.** We use IconQA (Lu et al., 2021) and ScienceQA (Saikh et al., 2022) as primary evaluation benchmarks, which represent challenging real-world scenarios without prior knowledge of data distribution where long-tailed biases manifest in various forms beyond simple class imbalance. ScienceQA includes 8K multimodal multiple-choice questions spanning 3 subject areas and 26 topics. IconQA contains 18K abstract diagram questions requiring icon-based reasoning, with entangled long-tailed biases in both visual concepts and question types. For zero-shot generalization evaluation, we employ OKVQA (Schwenk et al., 2022), OCRVQA (Mishra et al., 2019), GQA (Hudson & Manning, 2019), and TextVQA (Singh et al., 2019) as generalization test sets, following the established protocol in (Huang et al., 2024; Liang et al., 2025).

Implementation Details. Our experiments primarily utilize Qwen2.5-VL-7B (Team, 2025) as the large vision-language model (LVLM) backbone. To demonstrate the broad applicability and robustness of ATR, we also conduct experiments with LLaVA-1.5-7B (Liu et al., 2024a) for VQA tasks. For parameter-efficient fine-tuning (PEFT) methods, we employ LoRA (Hu et al., 2022) with a rank $r = 8$ and an alpha value of 16 on language model. To demonstrate the effectiveness of ATR, we also conduct experiments with DoRA (Liu et al., 2024b). LVLMs are trained for 3 epochs with a learning rate of $1e-4$. The batch size is set to 1 with gradient accumulation steps of 2 on each

Table 1. Comparison with state-of-the-art long-tailed recognition methods on ImageNet-LT and Place-LT. “SFT” means fine-tuning with cross-entropy loss. “Reweight” means sample reweighting based on the prior from (Du et al., 2023) when fine-tuning with cross-entropy loss. The optimal results are denoted by boldface. $\uparrow$ means improved accuracy compared with the sub-optimal results.

Methods	ImageNet-LT				Place-LT			
	Head	Medium	Tail	Overall	Head	Medium	Tail	Overall
<i>Training from scratch or a pre-trained backbone</i>								
cRT (Kang et al., 2019)	58.8	44.0	26.1	47.3	42.0	37.6	24.9	36.7
LWS (Kang et al., 2019)	57.1	45.2	29.3	47.7	40.6	39.1	28.6	37.6
MiSLAS (Zhong et al., 2021)	62.9	50.7	34.3	52.7	39.6	43.3	36.1	40.4
DisAlign (Zhang et al., 2021)	61.3	52.2	31.4	52.9	40.4	42.4	30.1	39.3
LiVT (Xu et al., 2023)	73.6	56.4	41.0	60.9	48.1	40.6	27.5	40.8
<i>Fine-tuning vision-language model</i>								
Zero-shot	60.8	59.3	58.6	59.8	37.5	37.5	40.1	38.0
SFT	74.4	56.9	34.5	60.5	50.8	38.6	22.7	39.7
BALLAD (Ma et al., 2021)	79.1	74.5	69.8	75.7	49.3	50.2	48.4	49.5
VL-LTR (Tian et al., 2022)	84.5	74.6	59.3	77.2	54.2	48.5	42.0	50.1
Candle (Shi et al., 2024)	76.3	69.5	62.9	71.2	46.9	45.4	44.5	45.8
MDPR (Jia et al., 2025)	79.4	74.0	70.0	75.5	50.3	51.4	50.9	50.9
SynBalance (Jiang et al., 2025)	81.9	75.8	68.7	77.5	51.2	50.2	49.9	50.5
<i>Fine-tuning large vision-language model</i>								
Zero-shot	65.4	62.6	60.4	63.4	51.5	32.8	28.0	38.6
SFT	84.9	76.7	64.7	78.2	54.7	39.7	33.0	43.8
Reweight (Du et al., 2023)	85.6	77.3	68.3	79.3	54.5	44.7	40.5	47.4
ATR(Ours)	90.2	82.8	71.5	84.1 $\uparrow_{4.8}$	58.3	49.2	43.5	51.4 $\uparrow_{0.5}$

Table 2. Comparison with state-of-the-art fine-tuning loss functions on ImageNet-LT and Place-LT. “Finetune” means the performance on fine-tuned dataset. “General” means the zero-shot performance on untrained dataset. “SFT” means fine-tuning with cross-entropy loss. The optimal results are denoted by boldface. $\uparrow$ means improved accuracy compared with the sub-optimal results.

Methods	ImageNet-LT								Place-LT							
	Finetune (Imageet-LT)				General (Place-LT)				Finetune (Place-LT)				General (Imageet-LT)			
	Head	Medium	Tail	Overall	Head	Medium	Tail	Overall	Head	Medium	Tail	Overall	Head	Medium	Tail	Overall
Zero-shot	65.4	62.6	60.4	63.4	51.5	32.8	28.0	38.6	51.5	32.8	28.0	38.6	65.4	62.6	60.4	63.4
<i>LoRA</i>																
SFT	84.9	76.7	64.7	78.2	44.7	26.5	24.3	32.6	54.7	39.7	33.0	43.8	57.2	57.0	54.6	56.8
MinorSFT	86.3	79.2	69.1	80.6	45.3	27.1	24.5	33.1	55.2	40.6	38.1	45.4	57.8	57.3	55.2	57.2
FisherSFT	87.4	78.5	68.3	80.5	45.1	27.6	24.8	33.3	56.9	43.8	37.1	47.2	58.3	58.4	55.7	58.0
ATR(Ours)	90.2	82.8	71.5	84.1 $\uparrow_{3.5}$	46.2	27.8	25.1	33.9 $\uparrow_{0.6}$	58.3	49.2	43.5	51.4 $\uparrow_{4.2}$	58.9	58.6	56.1	58.4 $\uparrow_{0.4}$
<i>DoRA</i>																
SFT	83.7	75.4	64.6	77.1	45.2	26.1	24.9	32.7	54.2	40.1	33.4	43.9	57.0	56.4	55.2	56.5
MinorSFT	86.1	77.5	68.7	79.6	45.2	26.5	25.3	33.0	56.2	44.6	35.7	47.0	58.2	57.1	55.6	57.3
FisherSFT	87.2	78.0	68.4	80.2	45.8	26.0	25.4	33.0	55.8	43.7	36.4	46.6	57.9	56.6	55.7	57.0
ATR(Ours)	89.5	81.9	70.8	83.3 $\uparrow_{3.1}$	46.1	27.5	25.7	33.8 $\uparrow_{0.8}$	57.8	48.2	42.1	50.5 $\uparrow_{3.5}$	58.4	57.6	56.0	57.7 $\uparrow_{0.4}$

device. All experiments are conducted on 16 Ascend 910B GPUs (65GB memory each) using PyTorch bf16 precision within LLaMA-Factory framework (Zheng et al., 2024). For ATR hyperparameters, we set the initialized filtering threshold $\tau = 0.8$, incremental update rate $r = 0.1 \mid \mathcal{D} \mid^{-1}$ where $\mathcal{D}$ is the training set size, $\gamma = 2$, maximum weight $\ell_{max} = 3$, and masking probability $m = 0.05$, k -th percentile of highest token entropy that $k = 0.2$ based on preliminary experiments.

Prompt and Evaluation. To evaluate the classification performance of LVLMS on long-tailed benchmarks, we employ the prompt “Your task is to classify the image into one of the following classes: class name 1, class name 2, ...,

class name n ” to prompt model to output only the class name. We then compute accuracy via exact string matching between the predicted and corresponding ground-truth class names. Followings (Liu et al., 2019), we report accuracy separately for three subsets of classes on both two long-tailed dataset ImageNet-LT and Place-LT: head classes (> 100 images), medium classes ($20 - 100$ images), and tail classes (< 20 images). For VQA tasks, we calculate multi-choice questions accuracy follows (Huang et al., 2024; Liang et al., 2025).

Table 3. Comparison with state-of-the-art fine-tuning loss functions on visual question answering task IconQA and ScienceQA, with four dataset (OKVQA, OCRVQA, GQA, TextVQA) to evaluate the zero-shot performance. “*Finetune*” means the performance on fine-tuned dataset. “*General*” means the average zero-shot performance on these four untrained dataset. “*Avg*” means the average performance of “*Finetune*” and “*General*”. “SFT” means fine-tuning with cross-entropy loss. The optimal results are denoted by boldface. ↑ means improved accuracy compared with the sub-optimal results.

Methods	IconQA							ScienceQA						
	OKVQA	OCRVQA	GQA	TextVQA	General	Finetune	Avg	OKVQA	OCRVQA	GQA	TextVQA	General	Finetune	Avg
Fine-Tuning with LLaVA-1.5-7B architecture														
Zero-shot	55.6	68.2	61.4	60.3	61.3	22.9	40.6	55.6	68.2	61.4	60.3	61.3	65.8	63.5
LoRA														
SFT	50.8	55.4	54.2	50.3	52.7	83.7	68.2	51.4	61.2	52.7	53.1	54.6	74.6	64.6
MinorSFT	52.1	51.2	53.8	51.0	52.0	85.5	68.8	52.0	61.7	53.8	54.8	55.6	74.2	64.9
FisherSFT	50.5	54.3	54.1	50.8	52.4	85.1	68.8	52.1	60.5	53.2	54.0	55.0	75.7	65.3
ATR(Ours)	52.4	55.2	55.7	52.3	53.9	87.9	70.9 ↑2.1	53.6	62.4	54.9	55.2	56.5	78.6	67.6 ↑2.3
DoRA														
SFT	49.6	53.8	54.7	49.7	52.0	82.8	67.4	50.7	60.2	53.6	52.4	54.2	74.8	64.5
MinorSFT	49.5	55.7	53.2	50.1	52.1	84.5	68.3	52.4	59.3	54.2	53.8	54.9	76.1	65.5
FisherSFT	49.8	54.3	56.2	50.7	52.8	85.2	69.0	51.0	60.7	53.2	54.3	54.8	75.3	65.1
ATR(Ours)	50.2	55.9	57.3	50.8	53.6	86.7	70.1 ↑1.1	53.9	60.8	54.9	55.3	56.2	78.2	67.2 ↑1.7
Fine-Tuning with Qwen2.5-VL architecture														
Zero-shot	51.9	46.3	53.1	85.7	59.2	78.2	68.7	51.9	46.3	53.1	85.7	59.2	72.5	65.8
LoRA														
SFT	45.1	41.7	48.3	79.4	53.6	92.4	73.0	43.2	42.1	47.8	80.6	53.4	91.1	72.3
MinorSFT	46.4	43.1	49.0	79.9	54.6	94.5	74.6	44.1	42.5	49.1	81.1	54.2	94.7	74.5
FisherSFT	46.3	43.5	49.7	80.5	55.0	94.8	74.9	45.6	41.8	48.7	80.8	54.2	93.4	73.8
ATR(Ours)	47.2	44.1	50.3	81.2	55.7	98.9	77.3 ↑2.4	45.8	43.6	49.8	81.6	55.2	97.0	76.1 ↑1.6
DoRA														
SFT	44.8	41.9	47.5	79.6	53.5	93.1	73.3	43.3	41.3	47.2	79.1	52.7	91.8	72.3
MinorSFT	45.1	42.5	48.5	80.4	54.1	95.7	74.9	44.8	43.1	47.6	79.4	53.7	94.3	74.0
FisherSFT	45.6	40.8	48.2	80.7	53.8	94.2	74.0	45.3	43.8	48.2	80.0	54.3	94.1	74.2
ATR(Ours)	46.8	43.2	49.5	81.6	55.3	98.1	76.7 ↑1.8	45.6	44.1	48.6	80.4	54.7	96.4	75.5 ↑1.3

4.2. Main Results

Long-Tail Image Classification Results. Tab.1 presents the comparison results of our proposed ATR based on Qwen2.5-VL-7B with direct fine-tuning using cross-entropy loss, prior reweighting, and previous state-of-the-art long-tailed recognition methods on the ImageNet-LT and Place-LT benchmarks. The results show that direct fine-tuning with cross-entropy loss yields limited performance gains on tail data due to the overfitting on head data. ATR effectively mitigates this problem, consistently outperforming the fine-tuning baseline across all metrics and datasets. Moreover, ATR significantly surpasses reweighting technique based on long-tailed data distribution priors. This demonstrates ATR’s efficacy in fine-tuning LVLMs under long-tailed data distributions. Specifically, on ImageNet-LT, ATR achieves an overall accuracy of 84.1%, which further approaches the previous state-of-the-art on full ImageNet (90.2%) under long-tailed data scenario, and substantially outperforms previous long-tailed techniques, representing a significant step forward in long-tail research. On Place-LT, ATR also demonstrates superior performance, achieving the best overall accuracy. Note that large vision-language model Qwen2.5-VL-7B works much less effectively than vision-language (e.g. CLIP) on medium and tail set in Place-LT under zero-shot scenarios, while drags down the performance of ATR in this case. And the head performance of

ATR largely outperforms previous long-tailed techniques, which have near performance on head, medium and tail groups, showing the potential of LVLMs-based methods in further improvement. Furthermore, most of the existing CLIP-based methods utilize additional data (web images or generated images), leading to an improved tail performance, which will be included in future work.

Tab.2 further extends our analysis by comparing ATR with two effective supervised loss functions from large language models, FisherSFT (Deb et al., 2025) and MinorSFT (Xie et al., 2024), across two PEFT methods LoRA and DoRA. Across both fine-tuning datasets and generalization datasets, ATR consistently outperforms FisherSFT and MinorSFT, demonstrating its superior ability to handle long-tailed data.

Visual Question Answering Results. Tab. 3 presents the comparison VQA results, showcasing ATR’s performance under complex long-tailed distribution on IconQA and ScienceQA benchmarks. We evaluate ATR using the LLaVA-1.5-7B and Qwen2.5-VL-7B architectures combined with two PEFT methods (LoRA and DoRA), and compare it against direct supervised fine-tuning with cross-entropy loss, as well as MinorSFT and FisherSFT baselines. For fine-tuning scenarios, ATR significantly improves the performance of fine-tuned dataset on all scenarios. Furthermore, we also present the performance on four generalization

Table 4. Ablation study on ImageNet-LT and Place-LT.

DFHT	ARLT	VTM	Head	Medium	Tail	Overall
<i>ImageNet-LT</i>						
	SFT		84.9	76.7	64.7	78.2
✓			88.4	80.7	67.5	81.8
	✓		87.6	80.1	68.4	81.3
		✓	86.9	80.3	69.8	81.4
✓	✓		89.2	79.9	69.2	82.0
✓	✓	✓	90.2	82.8	71.5	84.1
<i>Place-LT</i>						
	SFT		54.7	39.7	33.0	43.8
✓			56.9	45.8	37.4	48.1
	✓		56.4	46.1	37.1	48.0
		✓	56.1	44.5	39.5	47.6
✓	✓		57.5	47.9	39.2	49.6
✓	✓	✓	58.3	49.2	43.5	51.4

Table 5. Hyperparameters sensitivity analysis on ImageNet-LT.

Hyperparameter	Value	Head	Medium	Tail	Overall
τ	0.7	89.7	82.3	71.4	83.6
	0.8	90.2	82.8	71.5	84.1
	0.9	90.0	82.5	71.1	83.8
ℓ_{max}	2	90.4	82.5	70.6	83.9
	3	90.2	82.8	71.5	84.1
	4	89.6	82.4	71.0	83.6

VQA tasks, which include OKVQA, OCRVQA, GQA, and TextVQA, and our ATR also achieves superior performance in all scenarios except the OCRVQA in IconQA dataset fine-tuning with LLaVA-1.5-7B architecture. This consistent improvement against FisherSFT and MinorSFT further validates ATR’s effectiveness in addressing the complex, entangled long-tailed biases present in VQA scenarios. It demonstrates that our proposed ATR significantly enhances both the performance and generalization of LVLMs on generative tasks under long-tailed data.

4.3. Ablation Studies

We conduct a comprehensive ablation study for each component in ATR: Dynamic Filtering of High-confidence Tokens (DFHT), Adaptive Reweighting of Low-confidence Tokens (ARLT), and the Visual Token Mask (VTM) strategy. Tab. 4 presents the results on both ImageNet-LT and Place-LT datasets to show the contribution of each component, using direct fine-tuning with cross-entropy loss as the baseline. The combination of DFHT and ARLT can effectively reduce the overfitting on head data, and VPT can significantly enhance the long-tailed performance. These results confirm that all three components of ATR are crucial and complementary, working together to achieve state-of-the-art performance in long-tailed LVLM fine-tuning.

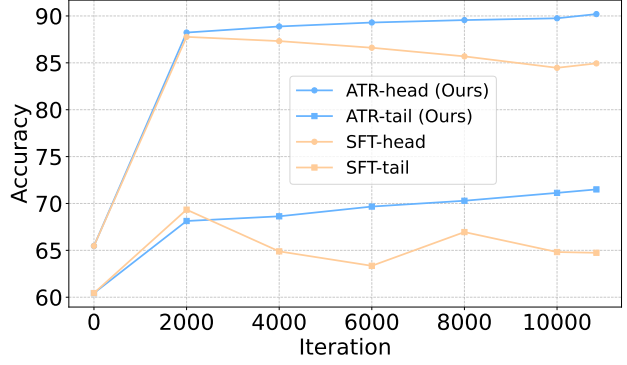


Figure 3. Training dynamics of head and tail data accuracy on ImageNet-LT between directly fine-tuning and our proposed ATR.

4.4. Discussion

Hyperparameter Sensitivity Analysis. The performance of ATR is influenced by several hyperparameters, particularly the dynamic confidence threshold τ and the maximum allowable loss value ℓ_{max} within bounded adaptive loss. Tab. 5 provides an sensitivity analysis of ATR to these hyperparameters on ImageNet-LT dataset, demonstrating that ATR is relative robust under different hyperparameters. Other hyperparameters sensitivity analysis detailed in Appendix Tab. 9.

Training dynamics. To evaluate the effectiveness of ATR in mitigating overfitting on head classes and improving long-tailed performance, we compare the training dynamics between direct fine-tuning with cross-entropy loss and our proposed ATR on the ImageNet-LT dataset. As illustrated in Fig. 3, direct fine-tuning achieves peak performance of head data at an early fine-tuning stage, after which it begins to degrade slightly, while tail class performance drops substantially. This demonstrates that direct fine-tuning rapidly overfits head data and significantly impairs tail class learning, failing to exploit their quantitative advantage. In contrast, our proposed ATR exhibits more balanced learning dynamics, with both head and tail data performance improving steadily throughout training. These results demonstrate that ATR provides a robust and effective strategy for enhancing long-tailed performance in LVLM fine-tuning.

5. Conclusion

To address the long-tail problem in large vision-language models (LVLMs) fine-tuning, we propose Adaptive Token Refinement (ATR), a novel framework to adaptively refine the learning process of LVLMs. ATR employs two complementary token-level operations: a bounded adaptive loss that dynamically filters the high-confidence tokens and reweights the low-confident tokens to mitigate overfitting on head data, and a visual token mask strategy that randomly

masks the low-entropy visual token to improve long-tail performance. Extensive experiments on large-scale long-tailed benchmarks and visual question answering tasks demonstrate that ATR consistently achieves significant improvements on both performance and generalization, enhancing long-tailed LVLs fine-tuning.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Conflict of Interest Disclosure

All authors disclosed no relevant relationships.

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A. Further Discussion

A.1. Threshold Strategy in Dynamic Filtering

To assess the effectiveness of our proposed dynamic threshold update mechanism within the Dynamic Filtering of High-Confidence Tokens component, we conducted a comprehensive comparative analysis against several alternative threshold scheduling strategies on the ImageNet-LT dataset. The strategies investigated include:

- **Fixed Threshold:** This baseline employs a constant threshold $\tau = 0.9$ throughout training, representing a static filtering criterion.
- **Cosine Threshold:** This strategy uses a cosine annealing schedule, progressively increasing the threshold from 0.8 to 0.98 along a cosine curve, offering gradual adjustment of filtering intensity.
- **ATR (Ours):** Our dynamic threshold update mechanism adaptively adjusts the threshold based on token prediction confidence, as defined in Eq. 5. This approach responds directly to the model’s evolving learning state.

As detailed in Tab. 6, our dynamic threshold update mechanism consistently demonstrated superior performance across all evaluated class groups (Head, Medium, and Tail). The fixed threshold, despite its simplicity, proved inadequate for adapting to the dynamic learning landscape, leading to suboptimal filtering. While the cosine threshold offered a more gradual adjustment than the fixed approach, it lacked the fine-grained adaptivity necessary to precisely respond to actual learning progress. In contrast, our ATR strategy automatically calibrates the threshold based on token-level confidence, enabling more effective filtering of well-learned head tokens while maintaining focus on challenging tail samples.

Table 6. Comparison of threshold strategies on ImageNet-LT.

Threshold Strategy	Head	Medium	Tail	Overall
Fixed Threshold	89.2	82.5	69.7	83.3
Cosine Threshold	89.4	82.0	70.2	83.2
ATR	90.2	82.8	71.5	84.1

A.2. Comparison of Filtered Tokens

To further reveal insights of our Dynamic Filtering of High-Confidence Tokens (DFHT) mechanism, we conducted a detailed analysis of the distribution of filtered tokens across different class groups during training on ImageNet-LT. Fig. 4 presents the percentage of filtered tokens originating from the head, medium, and tail classes, offering a comparative perspective against FisherSFT, which samples a subset of

the training dataset for informative training. The visualization reveals two key findings:

- **Head-biased filtering:** A significant majority of filtered tokens consistently originate from head classes. This supports our hypothesis that Large Vision-Language Models rapidly develop overconfidence in predicting tokens associated with frequently encountered head concepts. By filtering these high-confidence, well-learned tokens, ATR allows the model to allocate more learning capacity to less familiar concepts. Furthermore, our proposed ATR filters more head tokens compared to FisherSFT, demonstrating its effectiveness and fewer head bias.
- **Tail preservation:** Conversely, tokens from tail data are rarely filtered. This is crucial for mitigating the long-tail problem, where models often struggle to generalize to infrequent classes due to insufficient exposure. Moreover, our proposed ATR consistently preserves more medium and tail data compared to FisherSFT, ensuring better long-tailed learning.

This empirical evidence validates that DFHT successfully mitigates overfitting to head data, promoting a more balanced optimization.

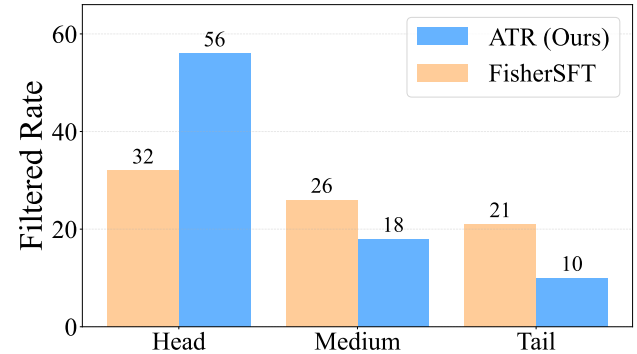


Figure 4. Distribution of filtered tokens across head, medium, and tail data during training on ImageNet-LT.

A.3. Visual Mask Strategy

The Visual Token Mask (VTM) strategy enhances model robustness by selectively masking low-entropy visual tokens. To demonstrate its effectiveness, we conducted a comparative study against several widely adopted data augmentation techniques on the ImageNet-LT dataset. The augmentation methods included:

- **Mixup:** This technique linearly interpolates between pairs of training images and their corresponding labels, generating synthetic samples to enrich the training data.

- **CutMix**: CutMix replaces rectangular regions of an image with patches from another training image, blending visual information and labels to encourage robust feature learning.
- **Random Token Mask**: This method randomly masks visual tokens without considering entropy, serving as a non-adaptive masking baseline.
- **VTM (Ours)**: Our proposed VTM employs an entropy-guided masking approach, selectively masking low-entropy tokens as defined in Eq. 10.

Tab. 7 presents the comparison results. While traditional augmentation methods like Mixup and CutMix offer moderate performance improvements, their image-level operation limits fine-grained, token-level control essential for optimizing Large Vision-Language Models. Random token masking, though token-level, showed limited effectiveness, especially for tail data, as it risks inadvertently masking informative high-entropy tokens crucial for distinguishing rare classes. In contrast, our entropy-guided VTM strategy achieved the best overall performance. By preserving high-entropy tokens rich in visual information and masking redundant low-entropy tokens, VTM effectively reduces the model’s reliance on simplistic visual cues. This fosters more robust feature learning, particularly beneficial for enhancing the generalization capabilities of tail data.

Table 7. Comparison of visual augmentation strategies on ImageNet-LT. “SFT” denotes fine-tuning with cross-entropy loss.

Augmentation	Head	Medium	Tail	Overall
SFT	84.9	76.7	64.7	78.2
Mixup	86.1	78.3	67.5	79.8
CutMix	85.6	78.8	67.9	79.9
Random Token Mask	86.5	79.2	67.3	80.4
VTM	86.9	80.3	69.8	81.4

B. Computation Cost

The VTM strategy requires computing token-level entropy via a forward pass. To mitigate the latency incurred by an additional serial forward pass within the current batch, we implement a memory bank for the storage of processed masks. This design enables the joint forward propagation of preprocessed masked samples and newly incoming samples in subsequent batches, leading to nearly negligible GPU memory overhead. The reason for this is that the newly added samples are detached from the computation graph and do not require backpropagation. Table 8 details the training time on ImageNet-LT using 16 Ascend 910B GPUs. Our analysis reveals that the full ATR framework, including entropy computation, adds reasonable training time given the substantial performance improvements.

Table 8. Computational cost comparison on ImageNet-LT with the same amount of training samples. “SFT” denotes fine-tuning with cross-entropy loss.

Type	Head	Medium	Tail	Overall	Train Time (h)
SFT	84.9	76.7	64.7	78.2	19.5
ATR	90.2	82.8	71.5	84.1	20

C. Extended Hyperparameter Sensitivity Analysis

Building upon the hyperparameter analysis in the main paper, we provide additional sensitivity studies for the remaining key hyperparameters: incremental update rate r , masking probability m , and the k -th percentile threshold for entropy-based masking. Tab. 9 presents comprehensive results on ImageNet-LT. The results demonstrate that our proposed ATR exhibits stable and robust performance across a reasonable range of hyperparameter values.

Table 9. Hyperparameters sensitivity analysis on ImageNet-LT.

Hyperparameter	Value	Head	Medium	Tail	Overall
r	0.05 $\mathcal{D}$ ⁻¹	89.7	82.3	71.2	83.6
	0.1 $\mathcal{D}$ ⁻¹	90.2	82.8	71.5	84.1
	0.2 $\mathcal{D}$ ⁻¹	90.1	82.1	70.8	83.6
m	0.02	89.3	81.6	70.9	83.1
	0.05	90.2	82.8	71.5	84.1
	0.1	89.9	82.6	71.0	83.8
k	0.1	89.5	82.1	71.3	83.5
	0.2	90.2	82.8	71.5	84.1
	0.3	89.8	82.6	71.4	83.8